

Quantum Numbers and Orbitals

Answer key with explanations

A. The quantum numbers

1. Meanings.

n: the shell (size and main energy). **l**: the subshell (shape). **m_l**: the orientation of the orbital in space. **m_s**: the spin of the electron (+ $\frac{1}{2}$ or $-\frac{1}{2}$).

2. $n = 4$.

(a) $l = 0$ (**4s**), 1 (**4p**), 2 (**4d**), 3 (**4f**). (b) Orbitals: $1 + 3 + 5 + 7 = 16 = 4^2$. Electrons: $2 \times 16 = 32 = 2 \times 4^2$.

B. Allowed or not?

3. Allowed sets.

- (a) **Allowed**: a 1s electron.
- (b) **Not allowed**: for $l = 1$, m_l can only be -1 , 0 or $+1$.
- (c) **Allowed**: a 3d electron.
- (d) **Not allowed**: m_s must be $+\frac{1}{2}$ or $-\frac{1}{2}$.
- (e) **Not allowed**: l must be at most $n - 1 = 2$.

4. Subshells.

(a) **2p**, 6 electrons (b) **3d**, 10 (c) **5s**, 2 (d) **4f**, 14

C. Electrons in atoms

5. Possible sets.

- (a) $1s^1$: (**1, 0, 0, $+\frac{1}{2}$**) (or $-\frac{1}{2}$)
- (b) Potassium is [Ar] $4s^1$: (**4, 0, 0, $+\frac{1}{2}$**) (or $-\frac{1}{2}$)
- (c) 2p: $n = 2$, $l = 1$, $m_l = -1, 0$ or $+1$, $m_s = \pm\frac{1}{2}$, e.g. (**2, 1, 0, $+\frac{1}{2}$**). (By Hund's rule, carbon's two 2p electrons are in different orbitals with the same spin.)

6. Pauli.

All electrons in one orbital share the same n , l and m_l , so they can only differ in m_s . m_s has only two values ($+\frac{1}{2}$ and $-\frac{1}{2}$), so a third electron would have to repeat a full set of quantum numbers, which the Pauli exclusion principle forbids.

D. Shapes and nodes

7. Shapes and nodes.

(a) s: a sphere centred on the nucleus. p_z : two lobes along the z axis, on either side of the nucleus, with a nodal plane (the xy plane) between them.

(b) 2s: **0 angular, 1 radial**. 2p: **1 angular, 0 radial**. 3d: **2 angular, 0 radial**.

E. Challenge

8. Block widths.

Each orbital holds 2 electrons, and each element in a block adds one more electron to that subshell. s has 1 orbital (2 electrons, **2 columns**), p has 3 (6 electrons, **6 columns**), d has 5 (10 electrons, **10 columns**) and f has 7 (14 electrons, **14 columns**). The numbers of orbitals come from m_l having $2l + 1$ values.